AQ5.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
The null space of a matrix $A_{4\times 3}$ A4×3 is

- the subspace $W = \{x \in R^4 \mid Ax = 0\}$W = {x ∈ R4 | Ax = 0} of R^4 R4.
- the subspace $W = \{x \in R^3 \mid Ax = 0\}$W = {x ∈ R3 | Ax = 0} of R^3 R3.
- the subspace $W = \{x \in R^2 \mid Ax = 0\}$W = {x ∈ R2 | Ax = 0} of R^2 R2.
- the first column of the matrix $A_{4\times 3}$ A4×3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
the subspace $W = \{x \in R^3 \mid Ax = 0\}$W = {x ∈ R3 | Ax = 0} of R^3 R3.

1 point
Nullity of a matrix $A_{3\times 4}$ A3×4 is

- 3
- 4
- the number of independent variables in the system of linear equations $Ax = 0$Ax = 0, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ x = [x1x2x3x4].
- the number of dependent variables in the system of linear equations $Ax = 0$Ax = 0, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$x = [x1x2x3x4].

No, the answer is incorrect.
Score: 0
Accepted Answers:
the number of independent variables in the system of linear equations $Ax = 0$Ax = 0, where $x = \begin{bmatrix} x_1 \\ x_2$

x = $\begin{bmatrix} \\ x1x2x3x4 \end{bmatrix}$.

Consider the system of linear equations $Ax = 0$Ax = 0, where $x = [x_1\ x_2\ x_3]^T$ x = [x1x2x3]T and

$A = \begin{bmatrix} 100 \\ 000 \\ 001 \end{bmatrix}$.A = $\begin{bmatrix} \\ 100000001 \end{bmatrix}$.

Use the above information to answer questions 3, 4 and 5.
1 point
Choose the correct statement from the following options.

- ○ x_1 x1, x_2 x2 and x_3 x3 all are dependent variables.
- ○ x_2 x2 is the only dependent variable.
- ○ x_1 x1 and x_3 x3 are the dependent variables and x_2 x2 is the only independent variable.
- ○ x_1 x1, x_2 x2 and x_3 x3 all are independent variables.

No, the answer is incorrect.
Score: 0
Accepted Answers:
x_1 x1 and x_3 x3 are the dependent variables and x_2 x2 is the only independent variable.
1 point
Which one of the following vector spaces represents the null space of AA?

- ○ $\{(t_1, 0, t_2) \mid t_1, t_2 \in R\}$ {(t1, 0, t2) | t1, t2 ∈ R}
- ○ $\{(0, t, 0) \mid t \in R\}$ {(0, t, 0) | t ∈ R}
- ○ $\{(t_1, t_2, 0) \mid t_1, t_2 \in R\}$ {(t1, t2, 0) | t1, t2 ∈ R}
- ○ $\{(0, 0, t) \mid t \in R\}$ {(0, 0, t) | t ∈ R}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(0, t, 0) \mid t \in R\}$ {(0, t, 0) | t ∈ R}
What will be the nullity of AA?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Find the nullity of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

1 point
Let SS denote the set of solutions of the homogeneous system of linear equations $Ax = 0$Ax = 0. Which of the following statements is (are) true?

- [ ] If x_1 x1 and x_2 x2 are in SS, then $x_1 - x_2$ x1 − x2 is in SS.
- [ ] If xx is in SS, then cxcx is in SS, for any $c \in R$c ∈ R.
- [ ] If x_1 x1 and x_2 x2 are in SS, then $\alpha x_1 + \beta x_2$ αx1 + βx2 is in SS, for any $\alpha, \beta \in R$α, β ∈ R.
- [ ] If xx is in SS, then $A^n x = 0$Anx = 0 for any $n \in N \setminus \{0\}$n ∈ N \ {0}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If x_1 x1 and x_2 x2 are in SS, then $x_1 - x_2$ x1 − x2 is in SS.
If xx is in SS, then cxcx is in SS, for any $c \in R$c ∈ R.
If x_1 x1 and x_2 x2 are in SS, then $\alpha x_1 + \beta x_2$ αx1 + βx2 is in SS, for any $\alpha, \beta \in R$α, β ∈ R.
If xx is in SS, then $A^n x = 0$Anx = 0 for any $n \in N \setminus \{0\}$n ∈ N \ {0}.

Find out the value of aa for which the matrix $\begin{bmatrix} 1 & 2 \\ 3 & a \end{bmatrix}$[132a] has nullity 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point

Level 2

1 point
Consider the coefficient matrix AA of the following system of linear equations:

$x_1 + x_2 + x_4 = 0$x1 + x2 + x4 = 0
$x_2 + x_3 = 0$x2 + x3 = 0
$x_1 - x_3 + x_4 = 0$x1 − x3 + x4 = 0

Which one of the following vector spaces represents the null space of AA?

- ○ $\{(t_1 + t_2, t_1, t_1, t_2) \mid t_1, t_2 \in R\}$ {(t1 + t2, t1, t1, t2) ∣ t1, t2 ∈ R}

○ $\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$ {(t1 − t2, −t1, t1, t2) | t1, t2 ∈ R}

○ $\{(t_1 + t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$ {(t1 + t2, −t1, t1, t2) | t1, t2 ∈ R}

○ $\{(t_1 - t_2, -t_1, t_1, -t_2) \mid t_1, t_2 \in R\}$ {(t1 − t2, −t1, t1, −t2) | t1, t2 ∈ R}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$ {(t1 − t2, −t1, t1, t2) | t1, t2 ∈ R}

Find the nullity of the matrix AA, where $A = [a_{ij}]$A = [aij] is of order $3 \times 3$3 × 3 and $a_{i,j} = \min\{i, j\}$ ai,j = min{i, j}, $i, j = 1, 2, 3$i, j = 1, 2, 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

1 point

Choose the correct set of options from the following.

☐ The nullity of a non-zero scalar matrix of order 3 must be 3.

☐ The nullity of a non-zero scalar matrix of order 3 must be 0.

☐ The nullity of a non-zero diagonal matrix of order 3 must be 3.

☐ The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The nullity of a non-zero scalar matrix of order 3 must be 0.

The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

Consider the system $Ax = b$Ax = b, where $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$A = [1428]. The number of solutions of when $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$b = [10] is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Check Answers